

is 473.6kbps for eight timeslots) in packet mode. The other main 2G system is code division multiple access (CDMA), which offers services similar to GSM but in different geographies.

Following the phenomenal and unexpected success of 2G, there was heightened anticipation of what 3G would bring, but 3G's contribution to date is a very mixed bag. On a technical level, 3G met the increased single-user data rates mandated by IMT-2000 requirements, but the uptake of much-hyped 3G services such as video telephony has been poor. This is why 3G subscriptions are just over seven per cent of 2G subscriptions, and many of these are primarily using telephony and basic messaging rather than 3G-specific data services.

Real uptake of data services did not start until the introduction of the so-called 3.5G packet based data services, like high-speed downlink packet access (HSDPA) on universal mobile telecommunication system (UMTS) and evolution data optimised (1xEV-DO) on CDMA2000. Dedicated data-only devices with flat-rate tariffs (for example, USB dongles used with laptops) are showing strong growth in some markets with good performance, at least for early adopters. 3G's impact is currently much less significant than the step from 1G to 2G.

Pre-4G technologies, such as mobile worldwide inter-operability for microwave access (WiMAX) and first-release 3G long-term evolution (LTE), have been available in the market since 2006 and 2009, respectively. Early WiMAX versions and first-release LTE captured the attention of wireless service provider giants around the world, but their market penetration is yet to be seen and their fate is not very clear.

4G refers to the fourth generation of mobile wireless standards. It is a successor to 3G and 2G families of standards. The expectation for 4G technology is basically the high-

quality audio/video streaming over end-to-end Internet protocol.

IMT-Advanced requirements

ITU-R for IMT-Advanced has set a specific level of minimum performance that each proposed technology needs to achieve to fulfill the objectives of IMT-Advanced. These requirements are demonstrated, at a glance, in Table I.

IMT-Advanced (4G) candidate technologies

Based on the thresholds set by ITU-R for IMT-Advanced, none of the wireless technologies at present, including LTE, high-speed packet access + (HSPA+) and WiMAX can be considered as 4G technology.

For HSPA+, main limitations are spectral efficiency, bandwidth and latency. On the latency front, HSPA+ cuts the user plane latency dramatically but it is still measured to be around 30ms to 40ms (round-trip delay) for best networks (refer Table I for comparison).

For LTE, being a much newer revolutionary technology, technology designers tried to improve the spectral efficiency by heavy use of multiple input multiple output (MIMO). LTE networks have been deployed with 2x2 MIMO from the start. This allowed the spectral efficiency to be boosted up; reaching peak spectral efficiency of 7.3bit/s/Hz (10MHz bandwidth), with the cell spectral efficiency level of 1.8bit/s/Hz (10MHz bandwidth). Even though LTE was defined for channelisation of 20MHz, 700MHz band, it is not large enough for such large bands to be assigned to a single operator. LTE user plane delay looks comparable to HSPA+ network.

WiMAX is fairly comparable to LTE in terms of spectral efficiency and delay capabilities since these share the same multiple access scheme and use similar modulation and coding scheme. Similar to LTE, WiMAX also uses 2x2 MIMO. WiMAX latency profile is similar to LTE due to

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